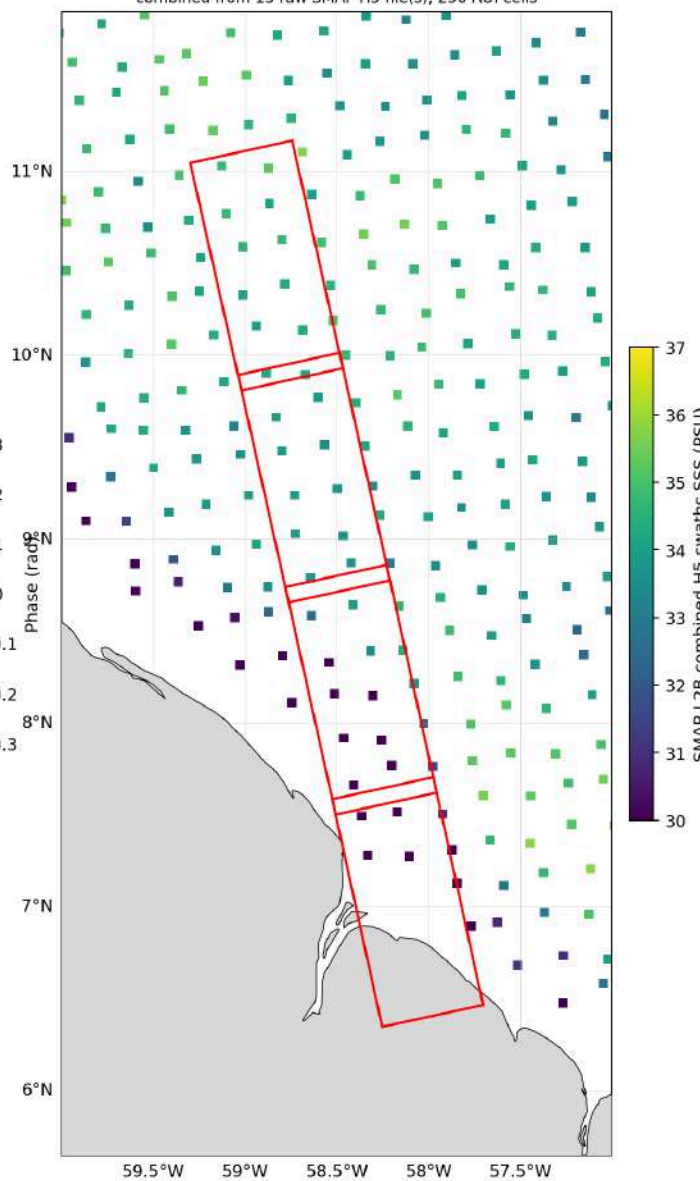


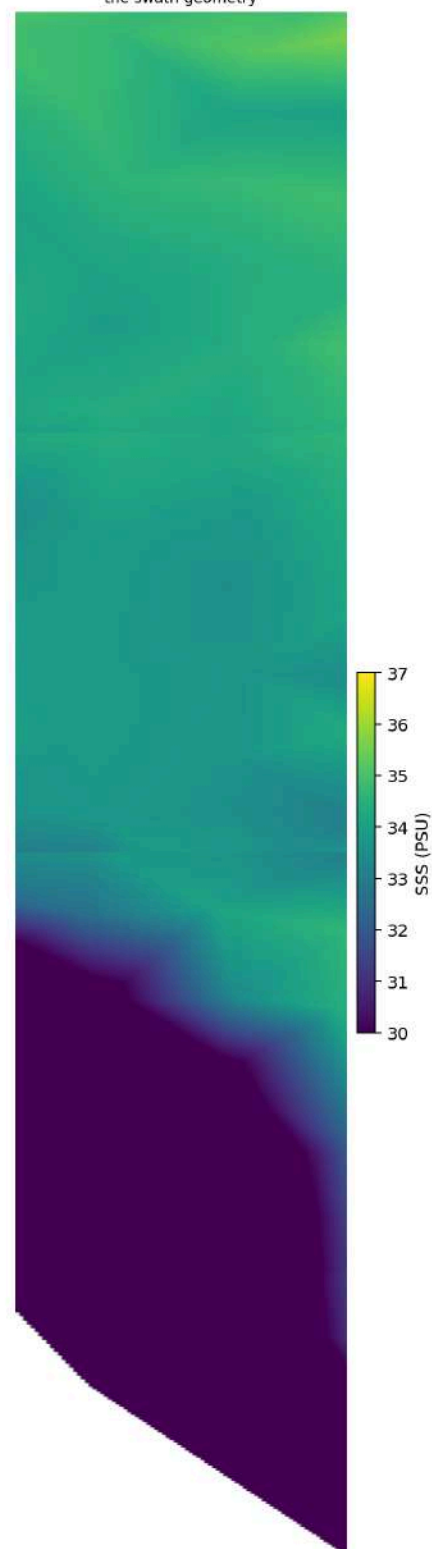
HH-VV phase difference (N up)
mean $+1.11^\circ \pm 1\sigma$ (20.64°)



SMAP L2B combined H5 swaths (unsmoothed swath points)
20.05–35.80 PSU (250 AOI cells)
combined from 13 raw SMAP H5 file(s); 250 AOI cells



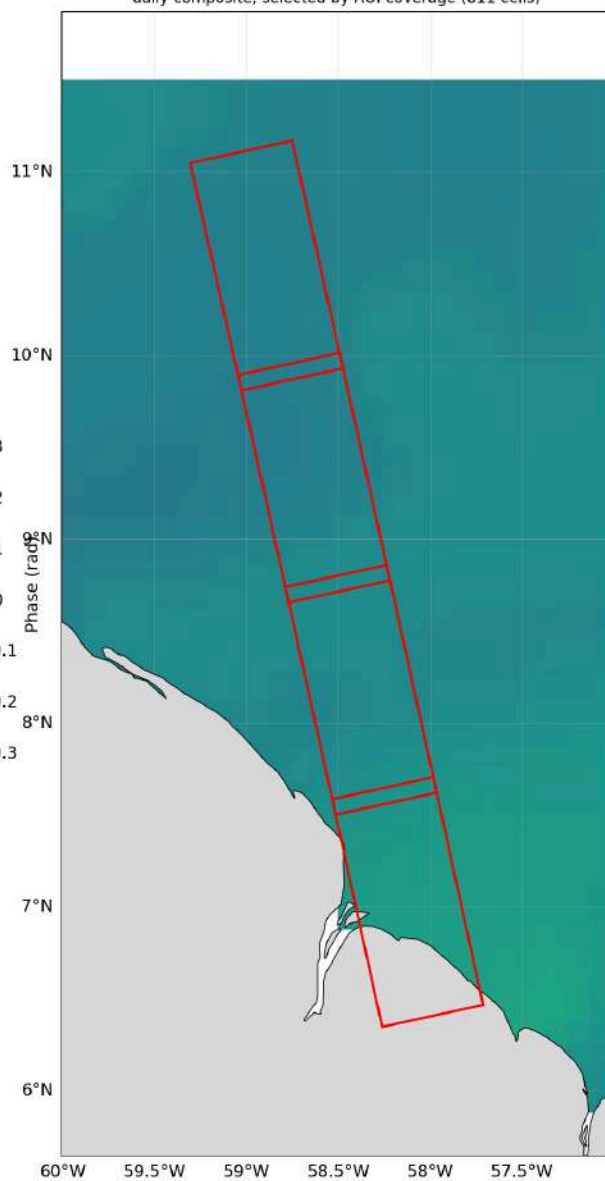
Salinity resampled onto
the swath geometry



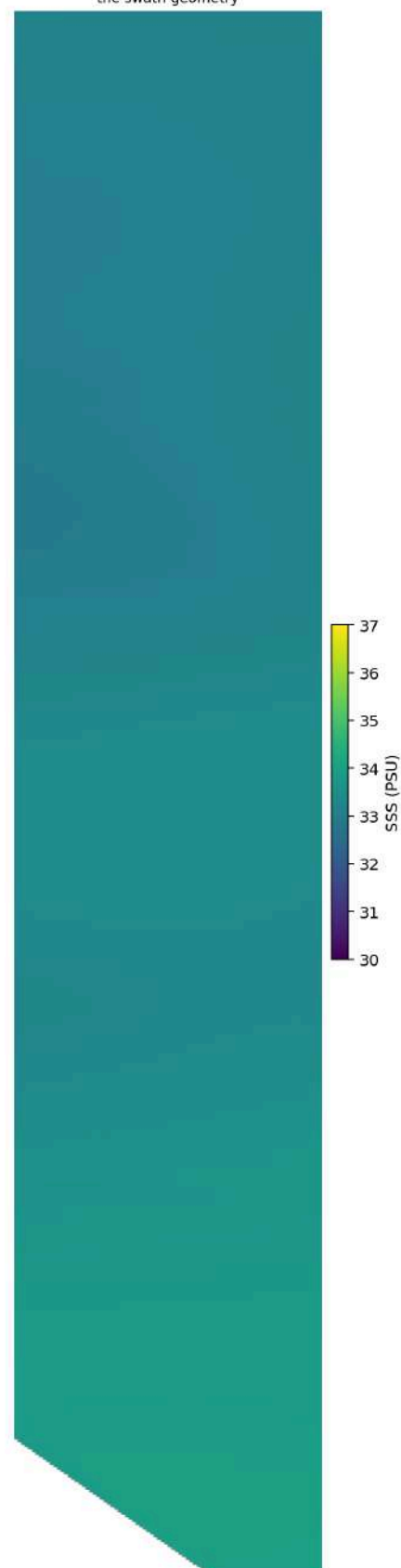
HH-VV phase difference (N up)
mean $+1.63^\circ \pm 1\sigma$ (20.19°)



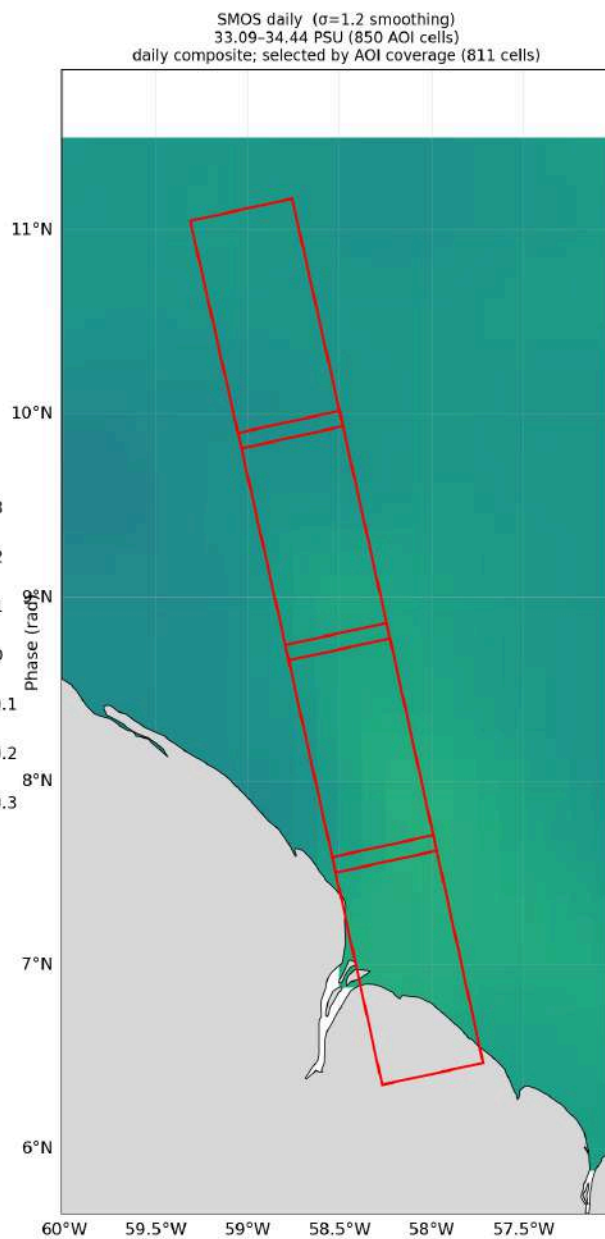
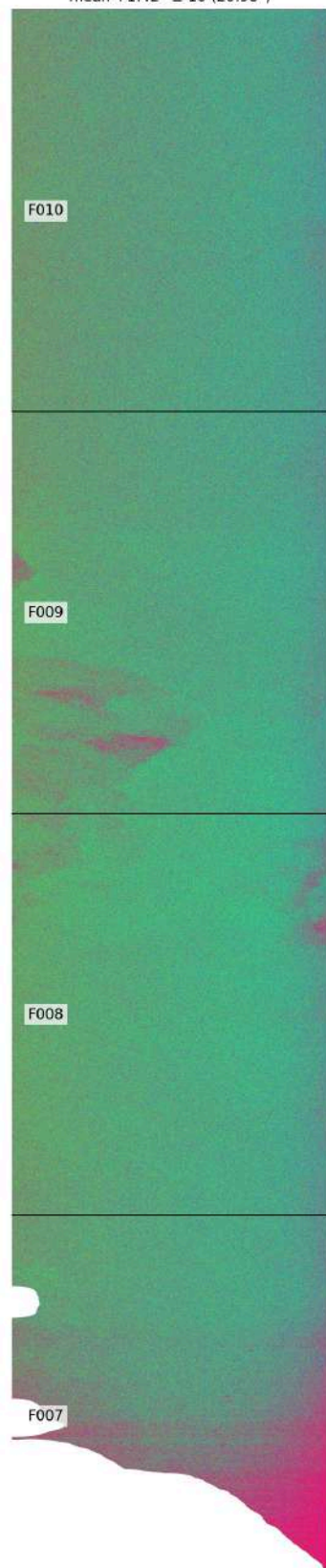
SMOS daily ($\sigma=1.2$ smoothing)
32.88-34.17 PSU (850 AOI cells)
daily composite; selected by AOI coverage (811 cells)



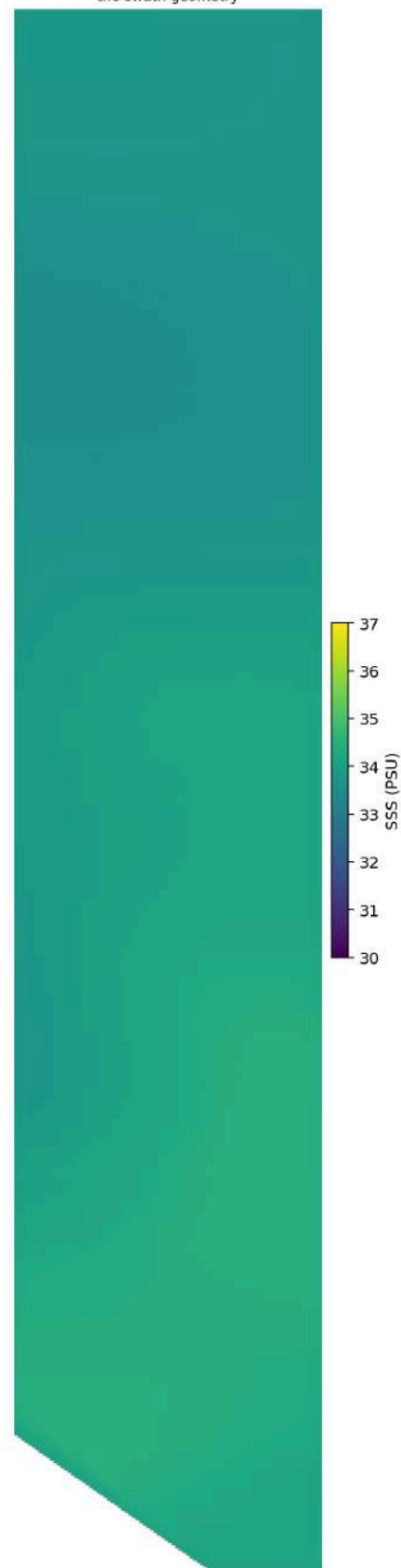
Salinity resampled onto
the swath geometry



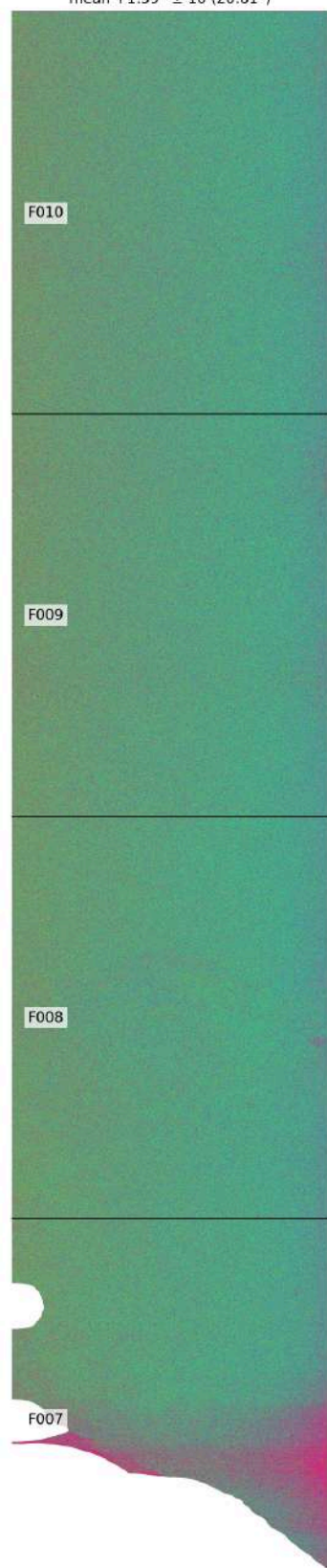
HH-VV phase difference (N up)
mean $+1.42^\circ \pm 1\sigma$ (20.98°)



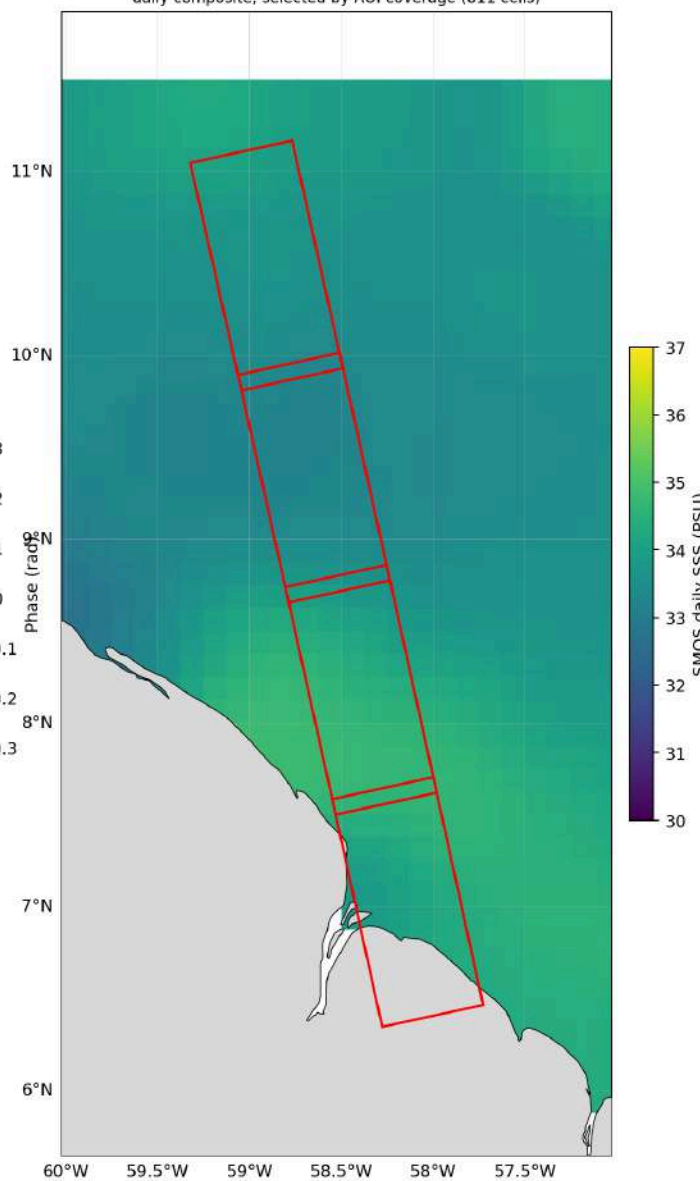
Salinity resampled onto
the swath geometry



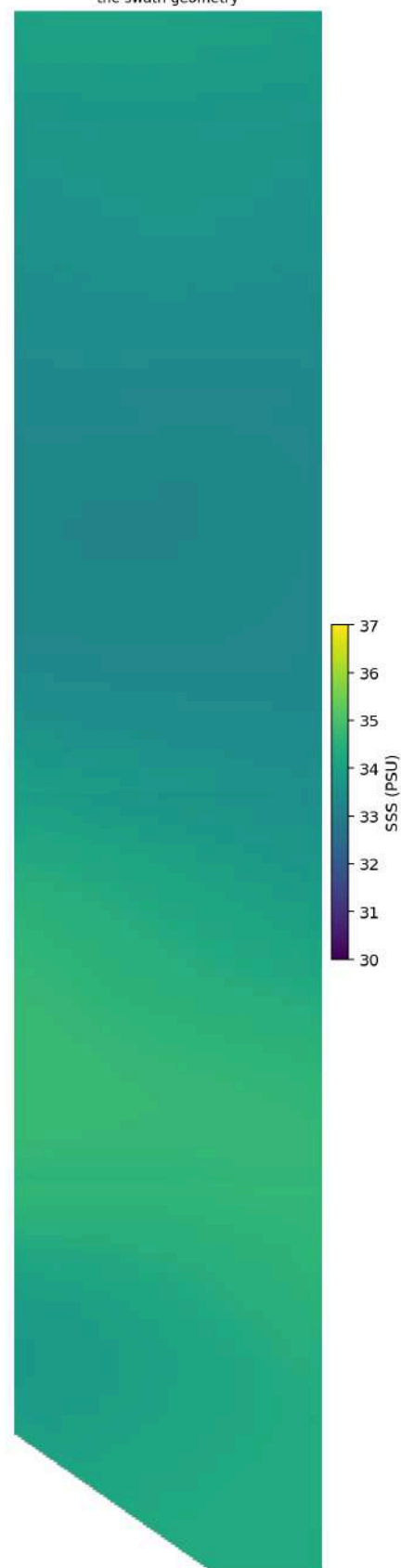
HH-VV phase difference (N up)
mean $+1.59^\circ \pm 1\sigma$ (20.61°)



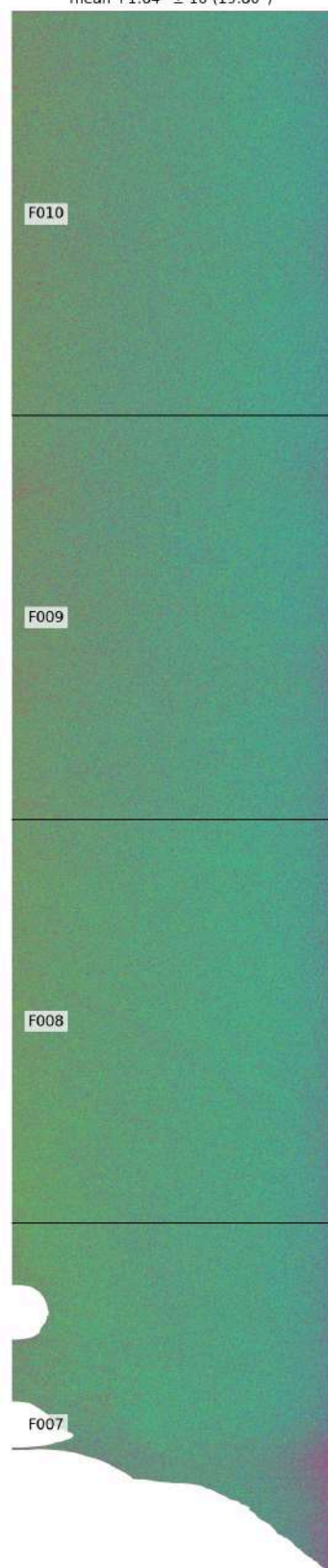
SMOS daily ($\sigma=1.2$ smoothing)
32.73–34.86 PSU (850 AOI cells)
daily composite; selected by AOI coverage (811 cells)



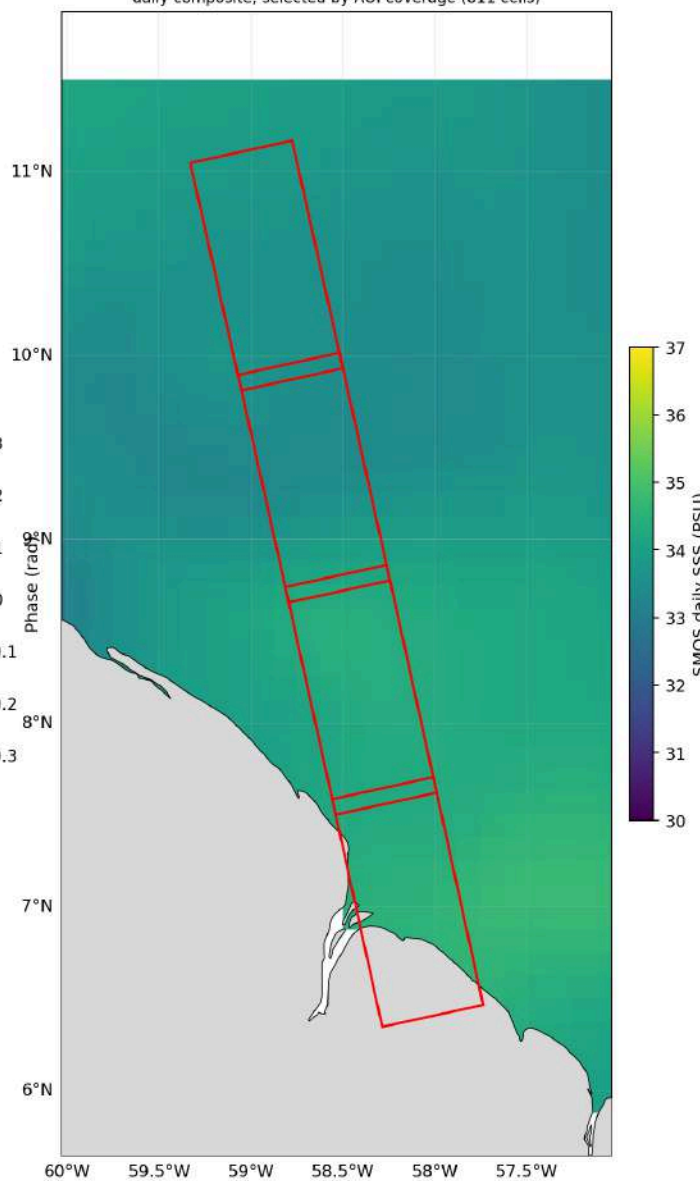
Salinity resampled onto
the swath geometry



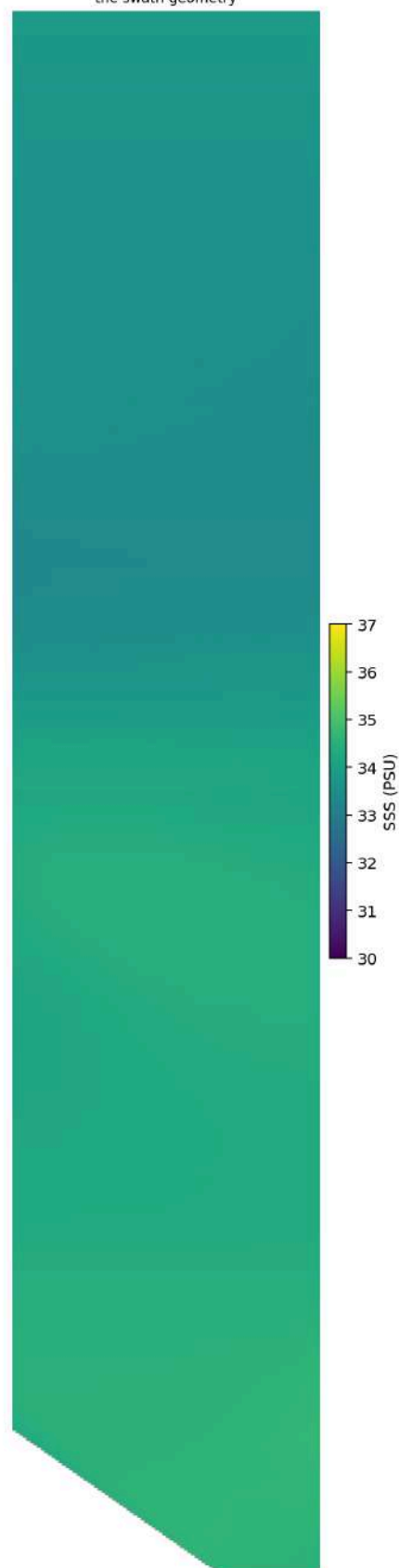
HH-VV phase difference (N up)
mean $+1.64^\circ \pm 1\sigma$ (19.80°)



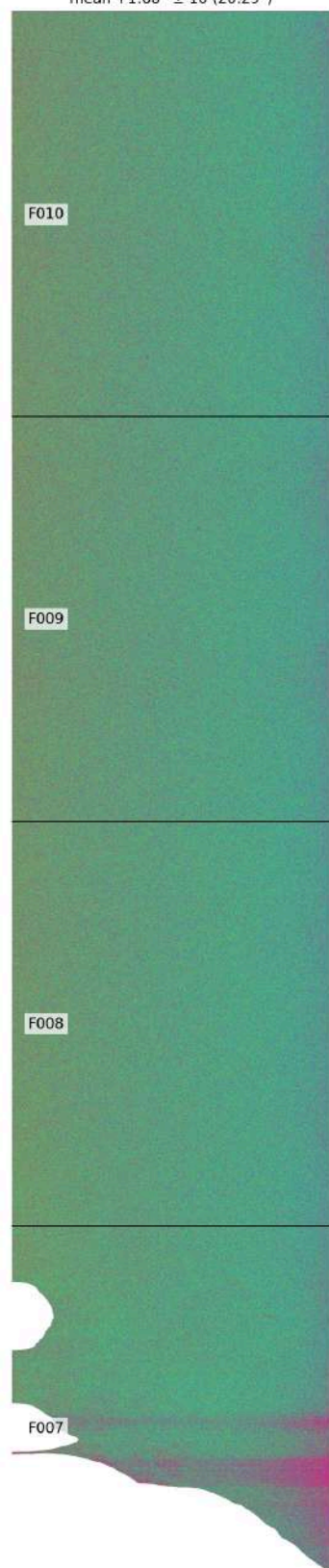
SMOS daily ($\sigma=1.2$ smoothing)
33.22-34.77 PSU (850 AOI cells)
daily composite; selected by AOI coverage (811 cells)



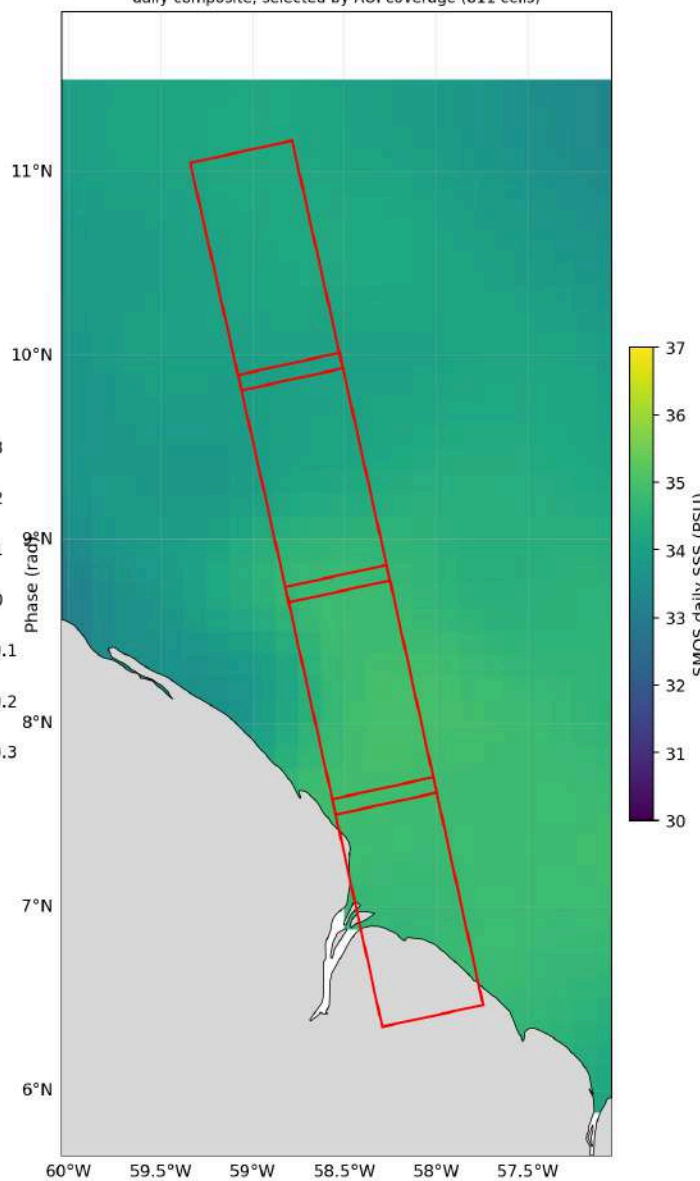
Salinity resampled onto
the swath geometry



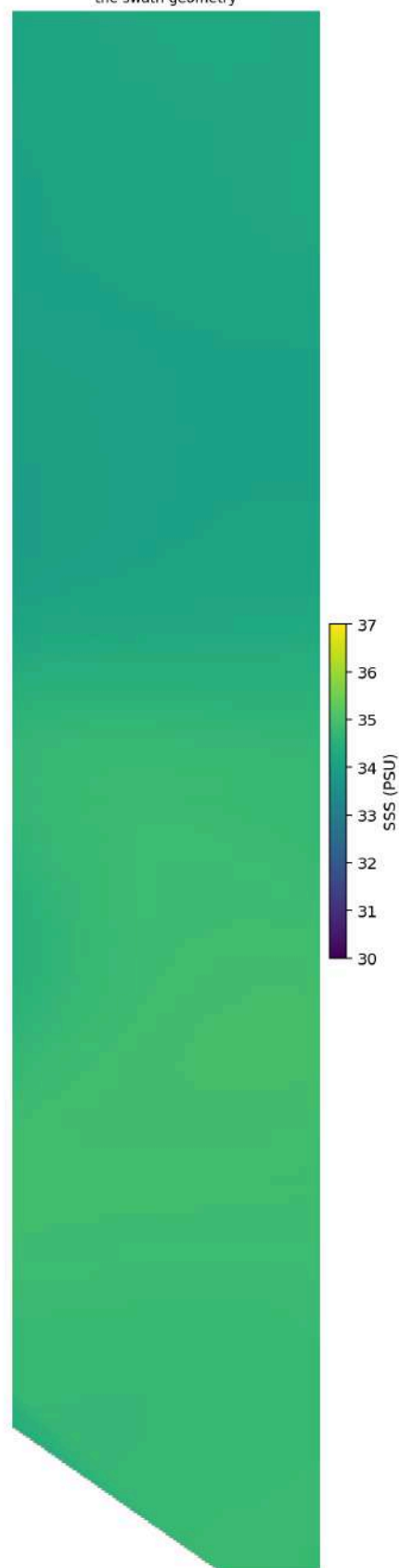
HH-VV phase difference (N up)
mean $+1.68^\circ \pm 1\sigma$ (20.29°)



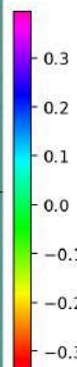
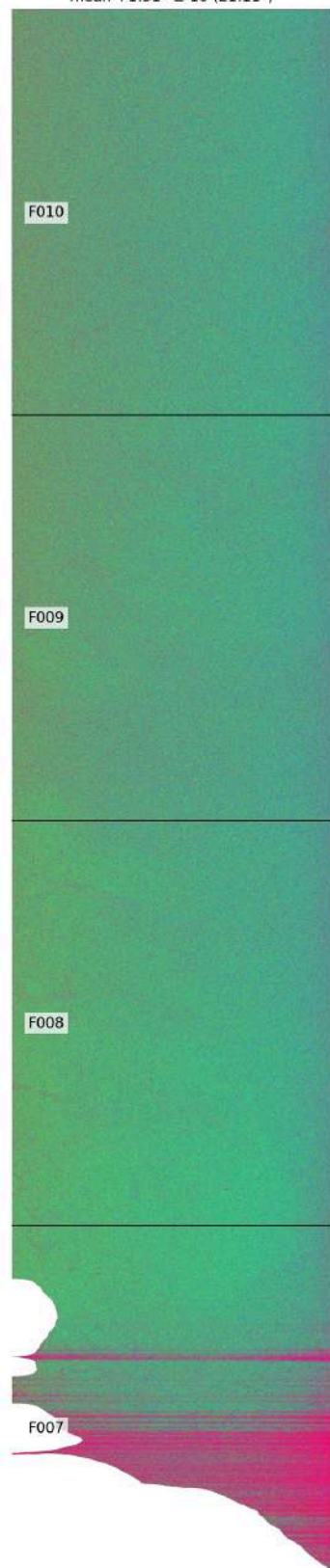
SMOS daily ($\sigma=1.2$ smoothing)
33.16–34.97 PSU (850 AOI cells)
daily composite; selected by AOI coverage (811 cells)



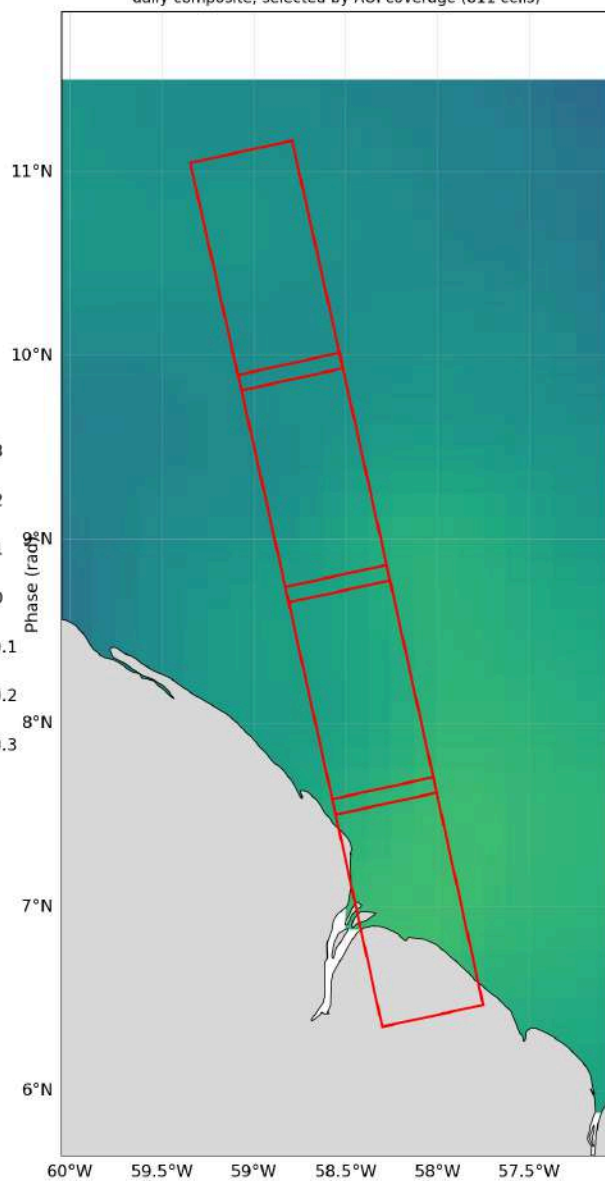
Salinity resampled onto
the swath geometry



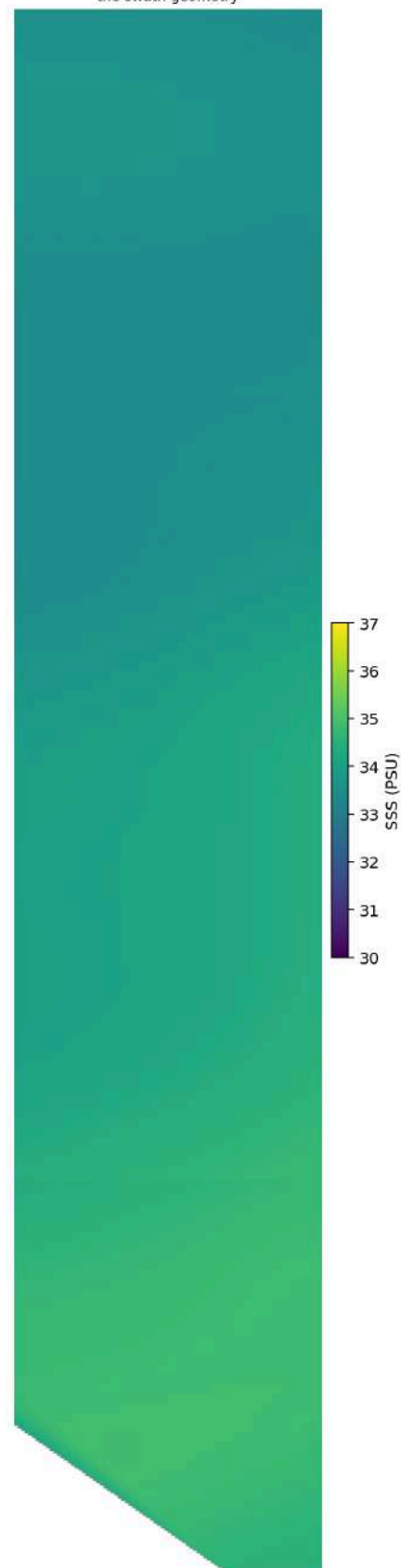
HH-VV phase difference (N up)
mean $+1.51^\circ \pm 1\sigma$ (21.11°)



SMOS daily ($\sigma=1.2$ smoothing)
32.47-34.91 PSU (850 AOI cells)
daily composite; selected by AOI coverage (811 cells)

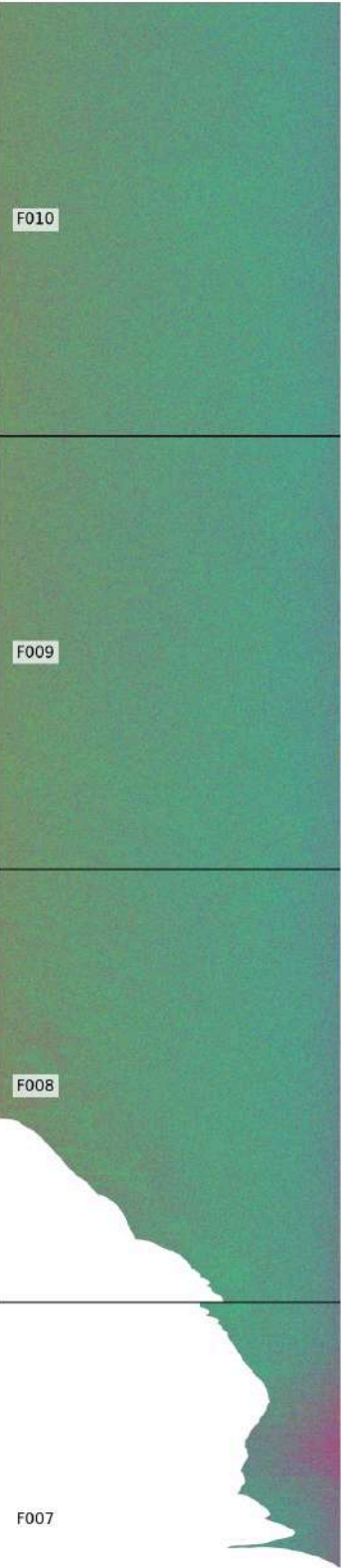


Salinity resampled onto
the swath geometry

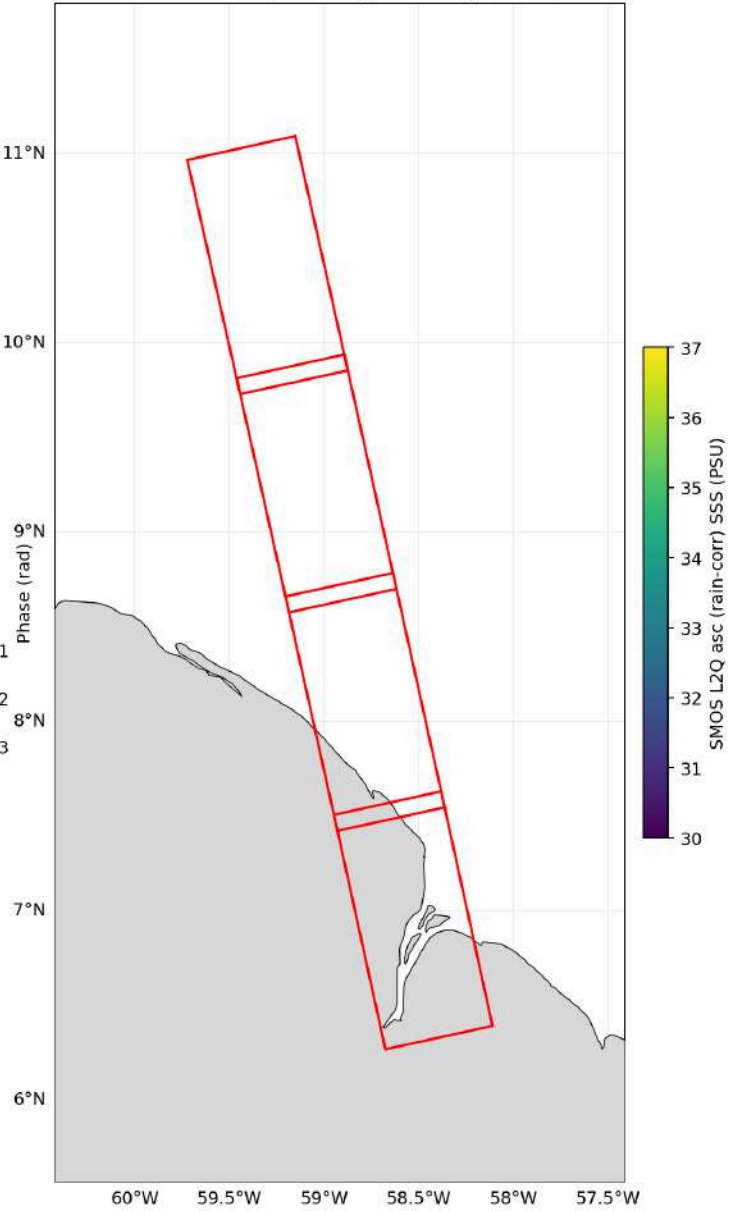


HH-VV phase difference (N up)
mean $+1.08^\circ \pm 1\sigma$ (22.20°)

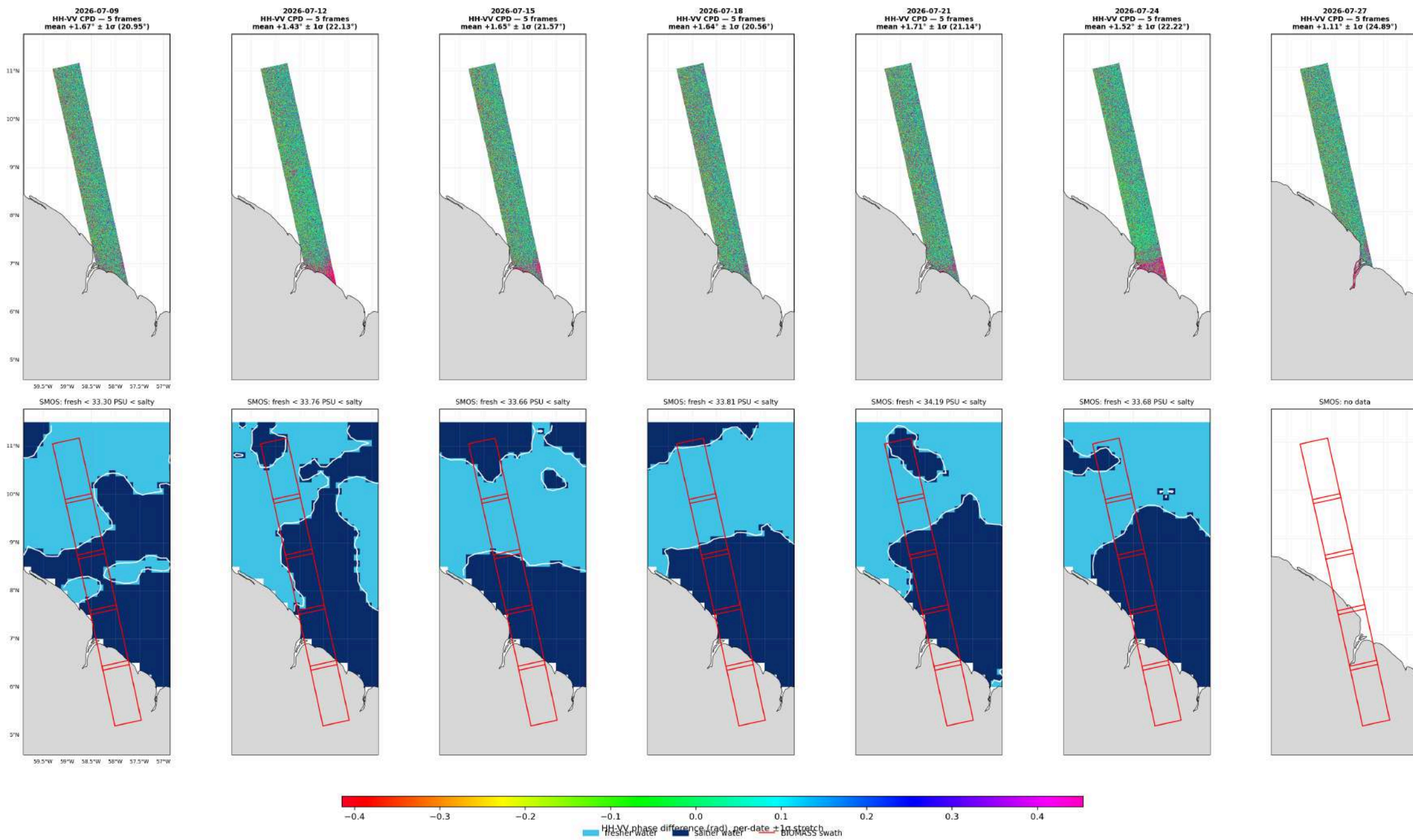
No salinity to resample

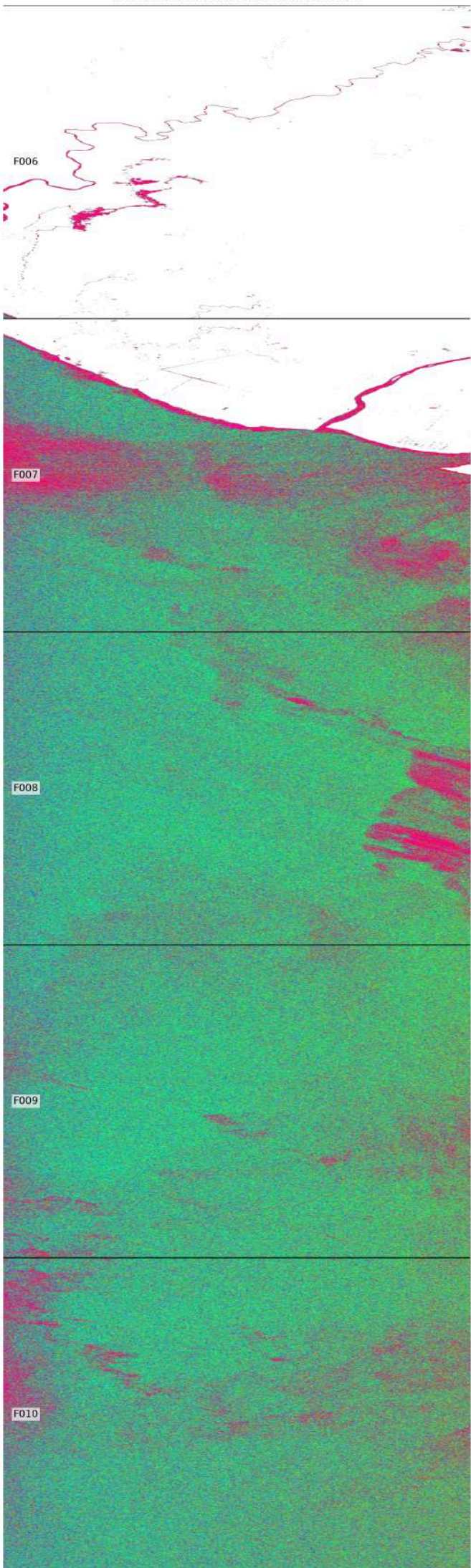


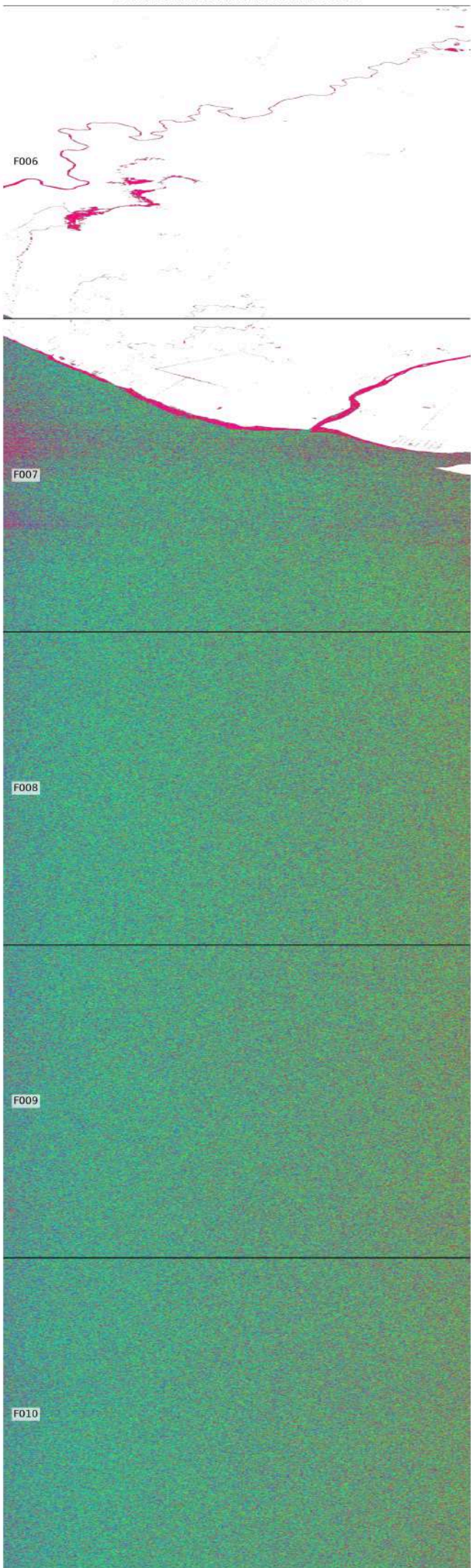
SMOS L2Q asc (rain-corr) ($\sigma=1.2$ smoothing)
no valid cells
+39 min from SAR; selected by AOI coverage (0 cells)

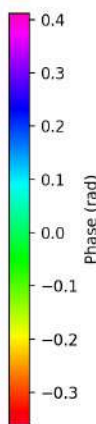
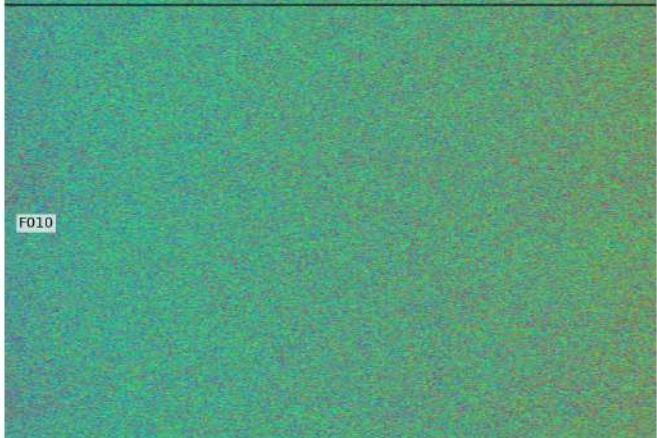
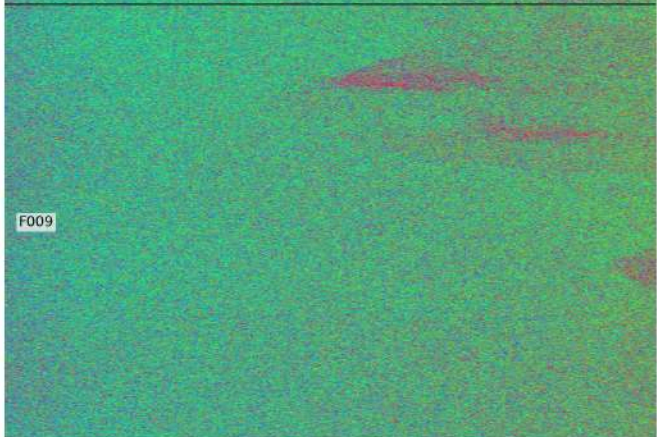
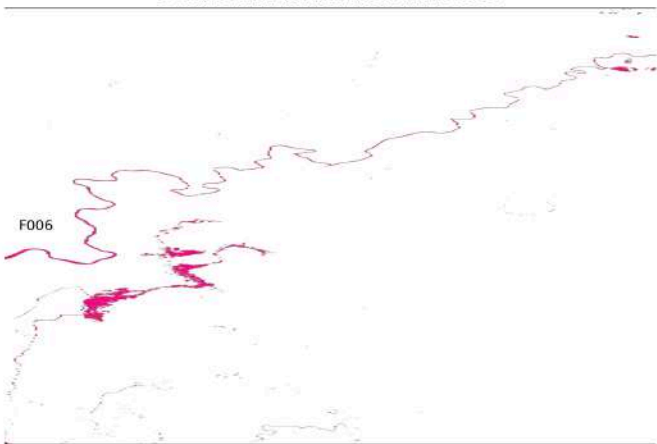


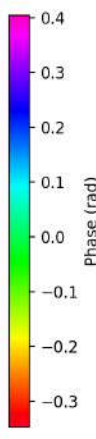
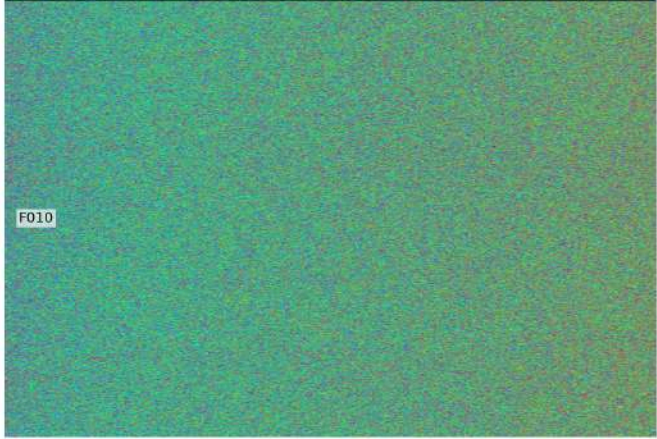
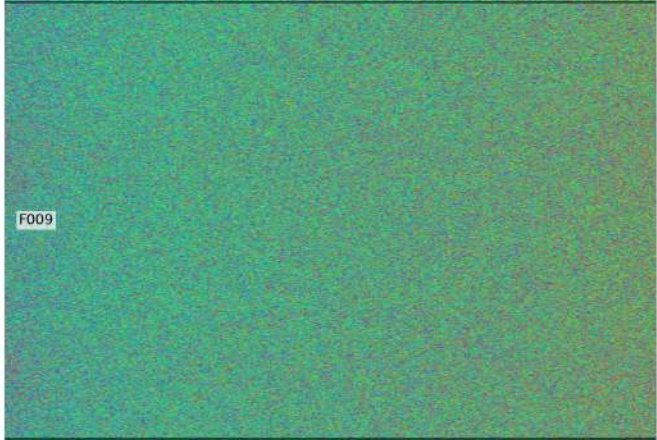
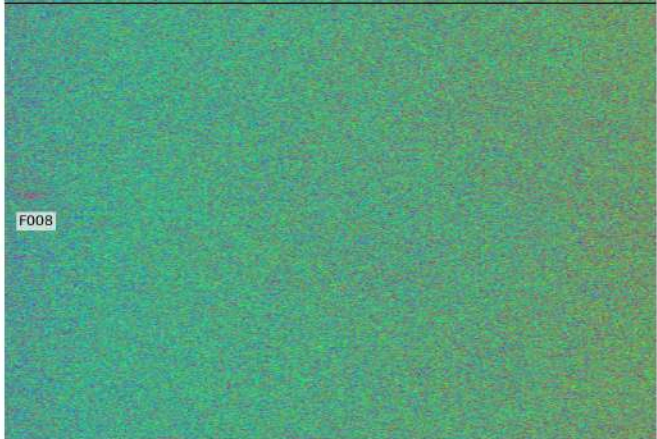
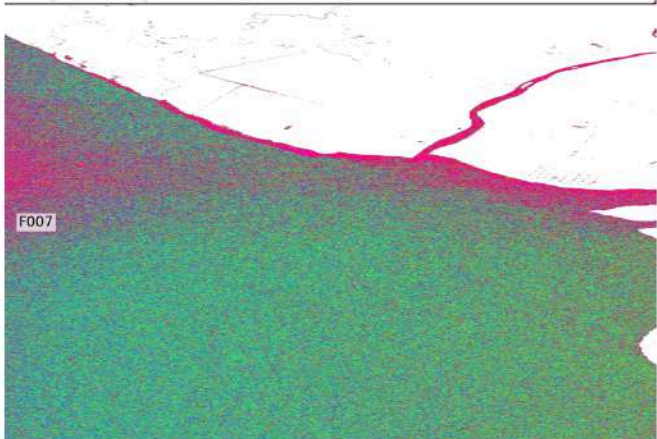
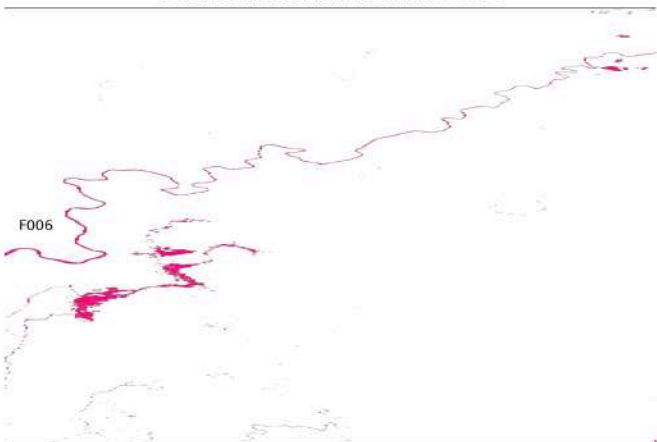
T036 Demerara — BIOMASS HH-VV phase difference (top) and SMOS fresh/salt classification (bottom)

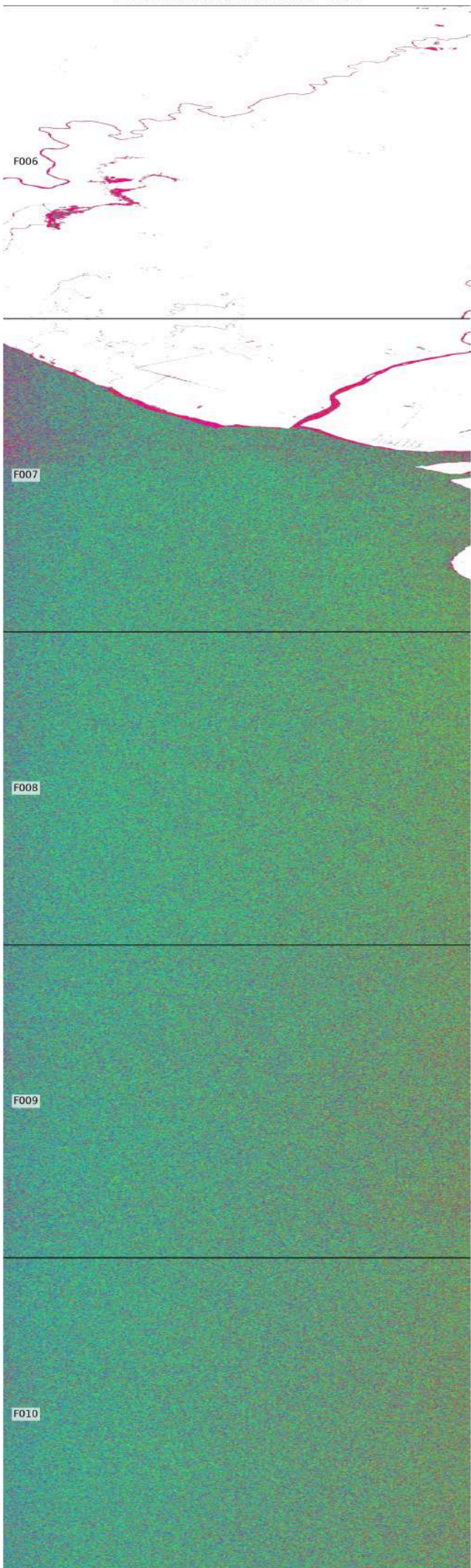


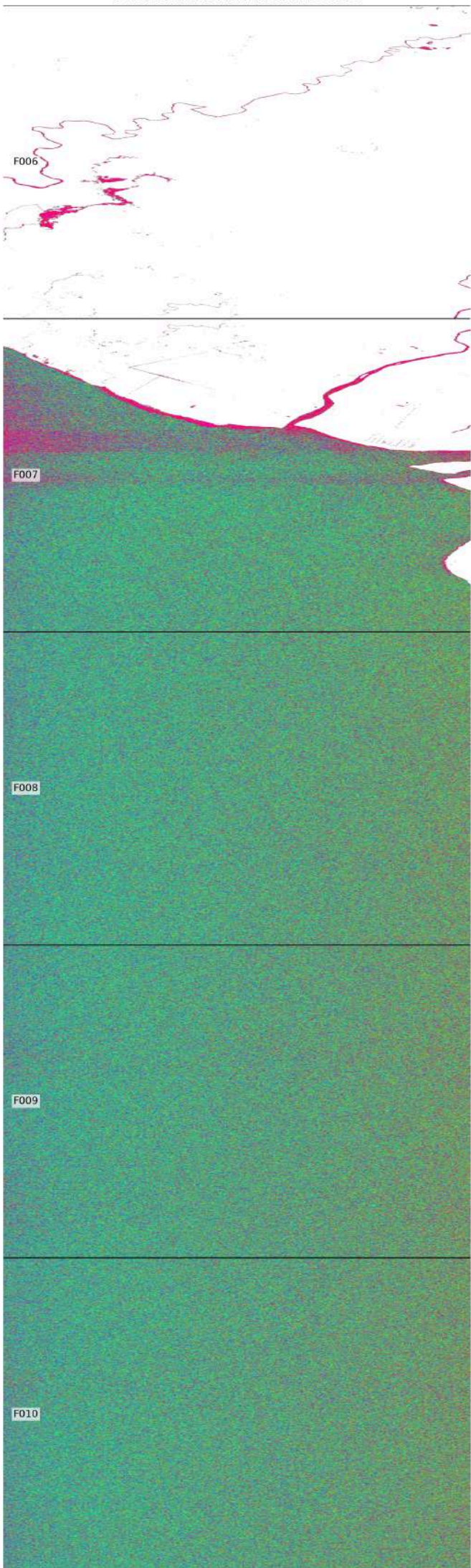


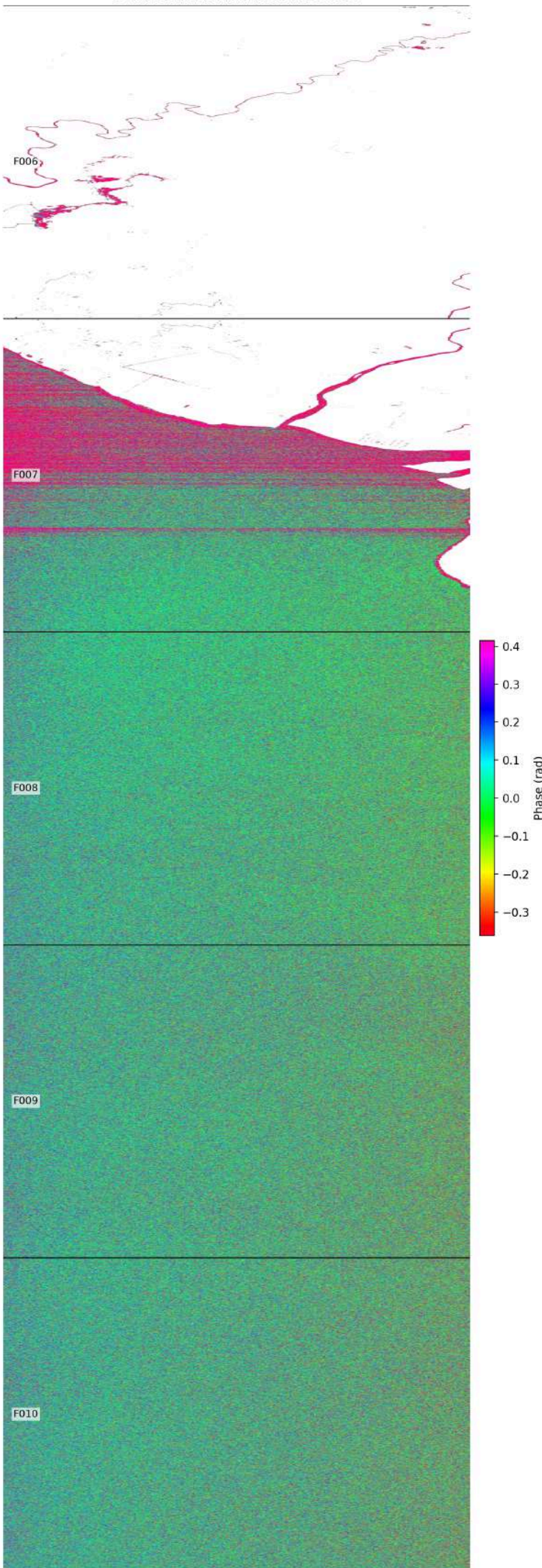


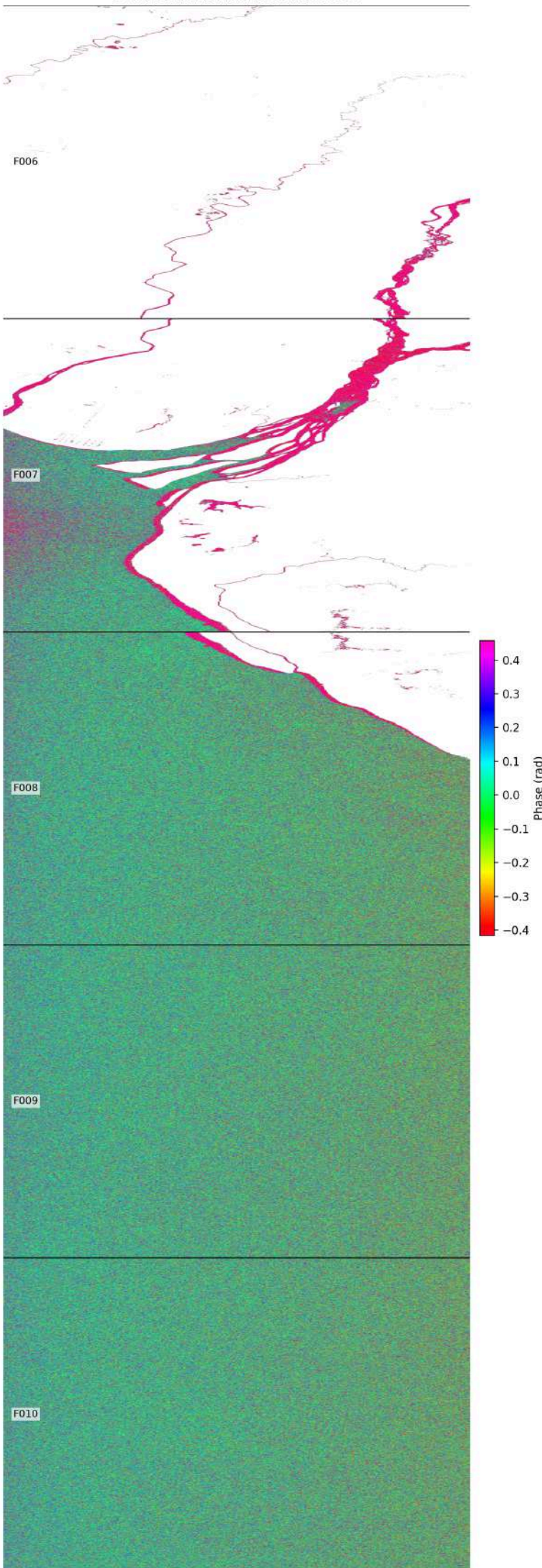




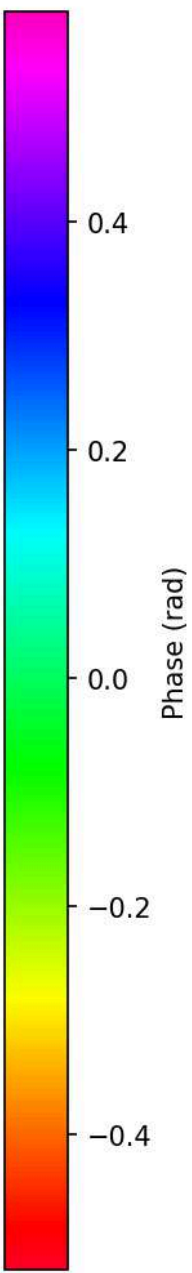




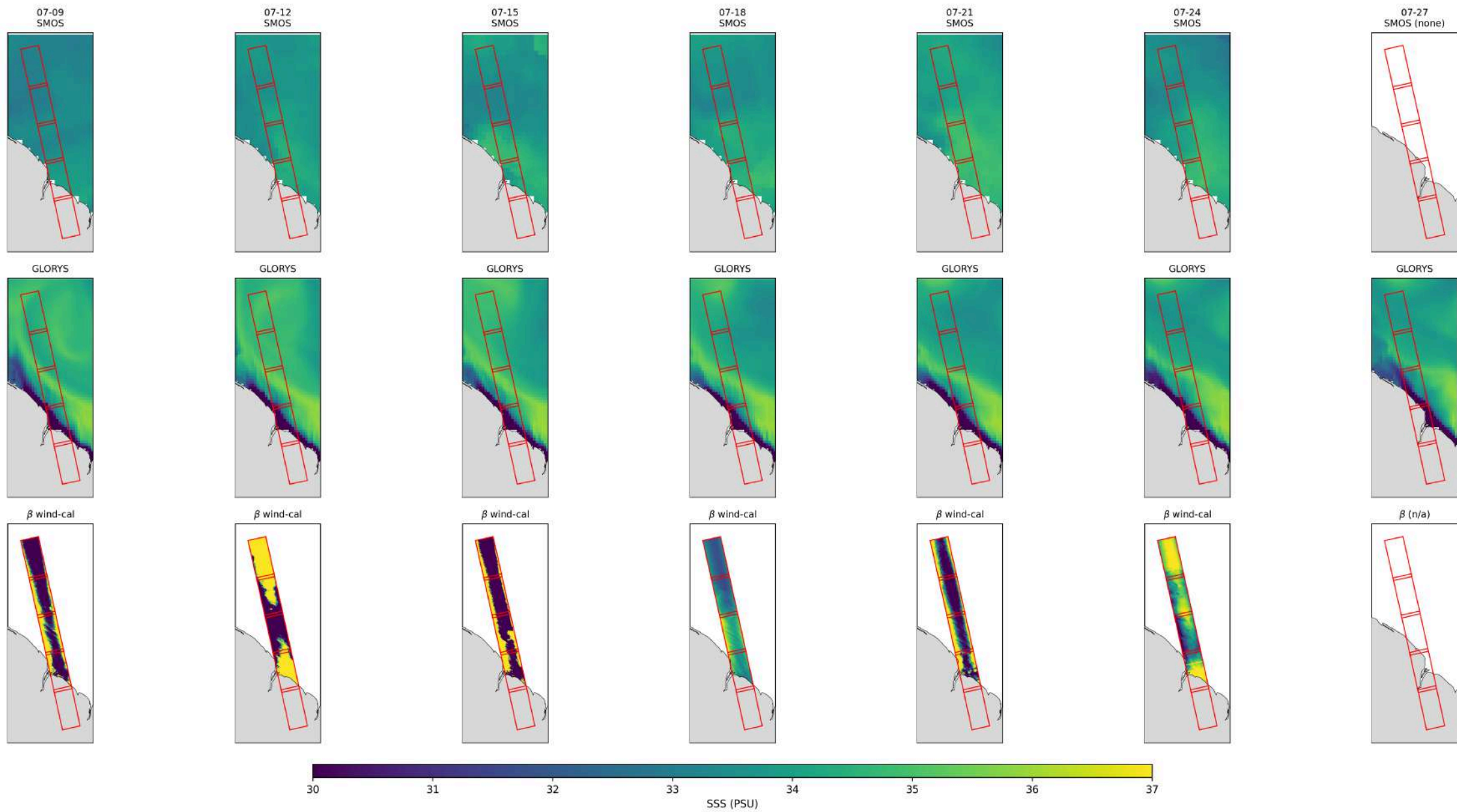




HH - VV Phase Difference — BIO_S1_SCS_1S_20260705T090254_20260705T090314_T_G01_M04_C01_T020_F310_02_DU3PUO
shift az-8 rg-24 (edge 1.77->1.95)



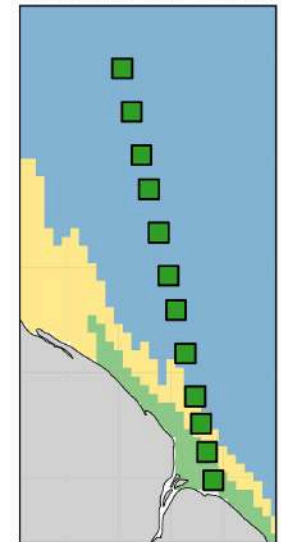
T036 Demerara — SMOS (top), GLORYS (middle), wind-calibrated β -derived SSS (bottom)
 red outline = BIOMASS swath; shared colour scale 30-37 PSU



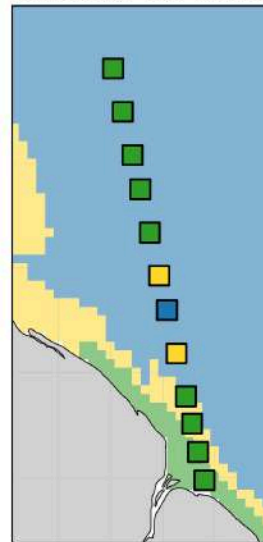
T036 Demerara — GLORYS salinity classes vs β -predicted classes, seven repeat passes

Calibration held fixed at 2026-07-06 values

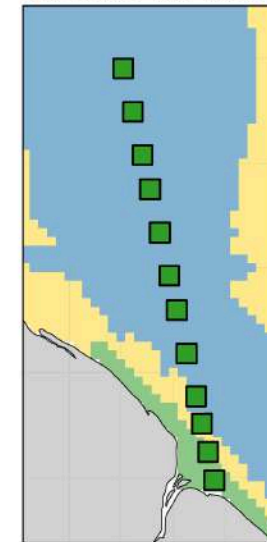
2026-07-09
wind 6.4 m/s bias -11.5 PSU



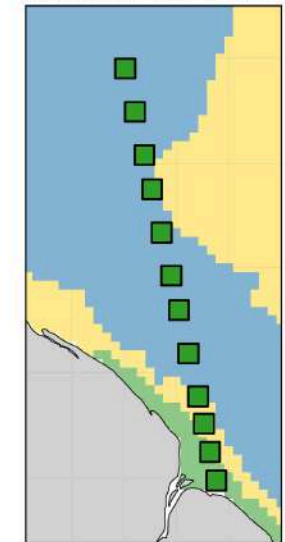
2026-07-12
wind 6.6 m/s bias -7.7 PSU



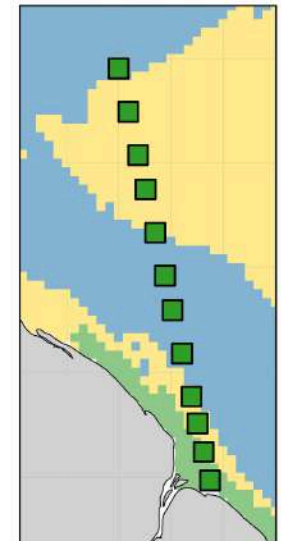
2026-07-15
wind 7.0 m/s bias -12.2 PSU



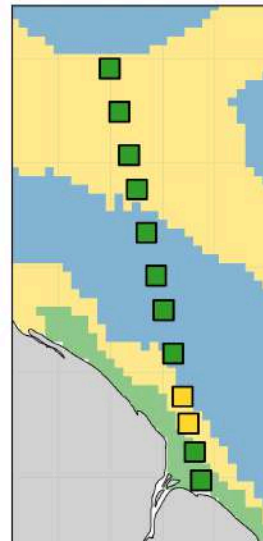
2026-07-18
wind 8.8 m/s bias -11.8 PSU



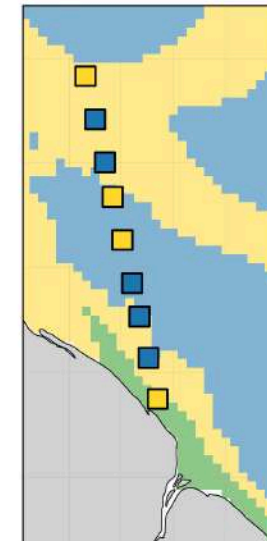
2026-07-21
wind 6.2 m/s bias -11.0 PSU



2026-07-24
wind 5.3 m/s bias -8.9 PSU



2026-07-27
wind 5.6 m/s bias +1.0 PSU



Fixed calibration from 2026-07-06:
 $|\beta| = 0.02798 \cdot SSS + 0.8638$

Applied unchanged to 7 later passes
(81 boxes):

pooled Spearman -0.006
pooled RMS 10.6 PSU
bias range -12.2 to +1.0 PSU

wind span 3.7–11.8 m/s
 β vs wind -0.502
 β vs salinity -0.006

Background = GLORYS class
Markers = β -predicted class

green fresher (<31) yellow mixed (31-34) blue saltier (>34)